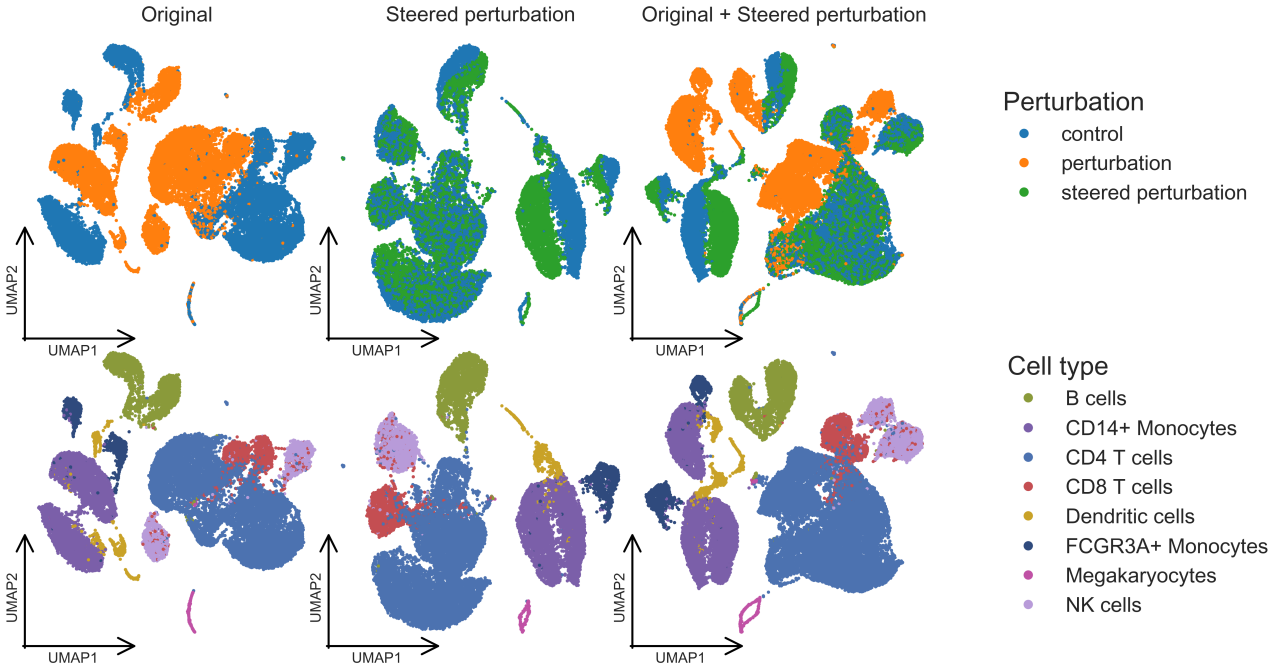

Figure 1: Steering IFN-β-perturbed PBMCs toward the control state. UMAP embeddings of PBMCs from the Kang et al. (2018) dataset. Each column shows a different combination of conditions: the original control and IFN-β perturbation (left), the control and steered perturbation (middle), and all three conditions overlaid (right). The top row is coloured by perturbation status and the bottom row by cell type, revealing that steered cells shift toward the control distribution while preserving cell-type structure.

Table 1: Pairwise Euclidean distances between stimulated (stim), control (ctrl), and steered stimulated (steered stim) cell populations per cell type. The mean is computed across all cells.

Cell Type	dist(stim, ctrl)	dist(stim, steered stim)	dist(ctrl, steered stim)
B cells	1.200	1.270	0.398
CD14+ Monocytes	2.716	2.419	1.191
CD4 T cells	0.934	0.827	0.280
CD8 T cells	0.814	0.793	0.182
Dendritic cells	4.268	3.760	1.457
FCGR3A+ Monocytes	3.148	3.347	0.995
Megakaryocytes	0.150	0.209	0.152
NK cells	0.963	0.931	0.297
Mean	1.38	1.31	0.37